

Characteristic properties of transition elements (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Recognise properties including variable oxidation state, high melting points, magnetic properties, catalytic properties, formation of coloured compounds and formation of complex ions with ligands.
- Explain why transition elements have variable oxidation states.
- Deduce the electron configurations of ions of the first-row transition elements.

Long before the development of our modern periodic table, it was known that the addition of certain metal salts during the production of glass would form coloured glass. You may have encountered coloured glass before in coloured glass bottles, artwork or even on windows. Metal salts containing transition elements are responsible for their colour. **Figure 1** shows an example of coloured glass and some transition elements and ions responsible for the apparent colour.



Figure 1. Coloured glass from transition elements.

Credit: Fred de Noyelle, [Getty Images](#)

(<https://www.gettyimages.com/detail/photo/heiligenkreuz-abbey-chapterhouse-stained-glass-royalty-free-image/538211406>)

When we looked at the formation of ions earlier in [section S2.1.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>), we identified transition elements as metals in the middle region of the periodic table that were capable of forming different ionic charges (**Figure 2**). But why are they called transition elements? What are they transitioning from? What makes these elements unique in comparison to other elements?

Figure 2. Region of the periodic table where transition metals are found.

 More information for figure 2

The image is a diagram of a periodic table highlighting the transition metals. The transition metals are marked by an oval outline encompassing the middle portion of the table, labeled as groups 3 to 12. It includes rows 4 to 7 horizontally, indicating the location where transition metals are typically found. The image features a grid-like structure representing the periodic table layout, with the label "Transition metals" inside the oval. The typical shape of the periodic table is maintained, showing different groups and periods as rows and columns. The elements themselves are not individually labeled, focusing instead on the general location of the transition metals within the table's framework. Asterisks are denoted below and to the left, likely indicating notation or additional groupings, though specifics are not provided in the image.

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Furthermore, when we looked at transition elements in [section S2.3.3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/transition-metals-hl-id-43565/>), we considered these as metals from groups 3–12 on the periodic table. This description we provided was incomplete. There is one group of elements that does not meet the criteria the International Union of Pure and Applied Chemistry (IUPAC <https://iupac.org/>) has set to define transition elements. Using [section 8](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/melting-points-and-boiling-points-of-the-elements-id-45110/>) of the DP chemistry data booklet, look at the melting points for groups 3–12 in period 4. Can you predict which element within the d-block of period 4 may not meet the criteria of a transition element?

A transition element is an element with an incomplete d-sublevel that is able to form cations with an incomplete d-subshell. They are called transition elements or transition metals because they exhibit transitional behaviour acting as somewhere

between s-block metals and p-block non-metals. The incomplete d-sublevels of transition elements give them characteristic properties.

If we consider the electron configuration of all the d-block elements in period 4 it becomes clear that the element zinc does not meet the criteria for a transition element. Zinc has the electron configuration $[\text{Ar}] 4s^2 3d^{10}$ and since it has a complete d-sublevel, it does not have the same characteristic properties as the other elements in this region of the periodic table. It is therefore not considered a transition element.

Characteristic properties of transition elements

1. Transition elements have variable oxidation states

Recall the formation of ions from transition elements from [section S2.1.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-ions-id-43120/>). Since transition elements have valence electrons occupying both s and d-sublevels, electron loss can occur from the outer d-sublevel, after loss from the outer s-sublevel occurs. This results in a variable number of electrons that are able to be easily removed from transition elements. Let's look at common ions of iron and their electron configurations in **Table 1**.

Table 1. Oxidation number and electron configuration for common species of iron.

Species	Oxidation number	Electron configuration
Fe	0	$[\text{Ar}] 4s^2 3d^6$
Fe^{2+}	+2	$[\text{Ar}] 3d^6$
Fe^{3+}	+3	$[\text{Ar}] 3d^5$

We will look more closely as to why transition elements have variable oxidation states in the next [section S3.1.9](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-variable-oxidation-states-id-44698/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/formation-of-variable-oxidation-states-id-44698/>).

2. High melting points

Recall from [section S2.3.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/>) that metallic bonding is the electrostatic attraction between a lattice of cations and delocalised electrons. We considered in [section S2.3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/>) how this strong attraction between cations in the lattice and delocalised valence electrons led to metals having high melting points. **Table 2** shows the melting points for period 4 metals. What do you notice about the melting points of metals from group 1 and 2 in comparison to the transition metals? What do you notice about the melting point of zinc in comparison to the transition metals?

Table 2. Melting point temperatures for period 4 metals.

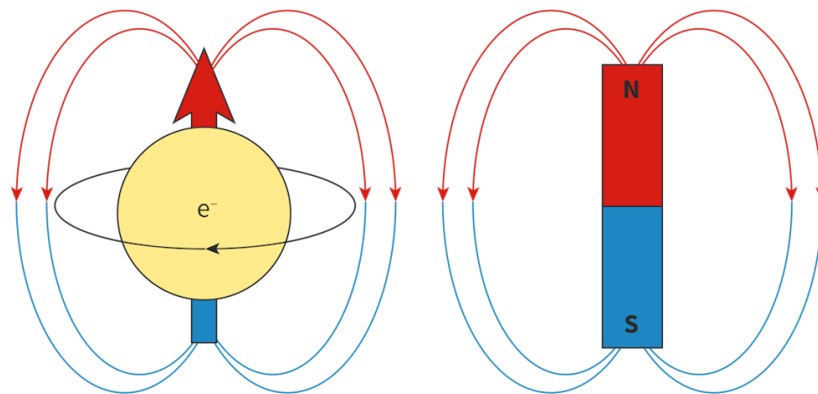
Element	Melting point (°C)
K	63.38
Ca	842.0
Sc	1541
Ti	1670
V	1910
Cr	1907
Mn	1246
Fe	1538
Co	1495
Ni	1455
Cu	1085
Zn	419.5

Since transition metals can form ions of varying oxidation states, this leads to a large number of delocalised valence electrons surrounding cations in the metallic lattice. The increased charge with a large number of valence electrons increases the strength of the metallic bond. These stronger metallic bonds require more energy to break, leading to higher melting points. This is why many everyday items you have encountered (e.g. cookware, coins, jewellery, etc.) are made from transition metals.

Since zinc can only form cations with a 2+ charge there are not as many delocalised valence electrons in comparison to transition metals. This results in a lower melting point of zinc in comparison to transition metals.

3. Magnetic properties

Did you know that electrons act as tiny magnets in atoms and molecules? The magnetic behaviour of electrons is the property called spin that you learned about in [section S1.3.5 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/). In that section, you learned that two electrons can occupy the same orbital, provided they have opposite spin. When elements have sublevels with unpaired electrons, magnetism can be observed in the substance. Since transition metals and ions have at least one unpaired electron in the d-sublevel, these metals and ions form a weak interaction with a magnetic field (which is called paramagnetism). This is shown in **Figure 3**.



Electron spin can be considered similar to a small magnet.

More information

The image shows a comparison between electron spin and a small magnet. On the left side, a diagram displays an electron represented by a yellow circle labeled with 'e-', surrounded by arrows indicating its spin and magnetic field lines in blue and red. The right side shows a bar magnet, divided into red and blue sections indicating the north (N) and south (S) poles, respectively, with field lines similar to the electron's. The caption reads, "Electron spin can be considered similar to a small magnet."

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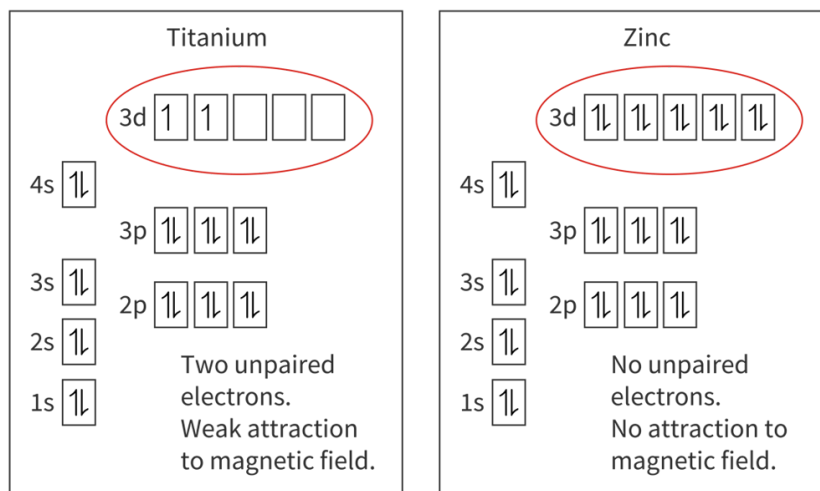


Figure 3. Unpaired d-electrons responsible for the magnetic behaviour of transition metals.

More information for figure 3

The image shows electron configuration diagrams for two transition metals, titanium and zinc, illustrating their magnetic behavior. Both diagrams include the distribution of electrons in various shells and subshells (1s, 2s, 2p, 3s, 3p, 3d, 4s).

On the left side, the diagram for titanium displays the 3d subshell highlighting the presence of two unpaired electrons. Text under the diagram notes, "Two unpaired electrons. Weak attraction to magnetic field," indicating that these

unpaired electrons contribute to titanium's magnetic properties.

On the right side, the zinc diagram shows a fully filled 3d subshell with no unpaired electrons. The accompanying text states, "No unpaired electrons. No attraction to magnetic field," explaining why zinc does not exhibit magnetic properties.

Overall, the image explains how the presence or absence of unpaired d-electrons in these transition metals affects their magnetic characteristics.

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Some of the transition metals form a metallic lattice structure where there is long range alignment of these unpaired electrons. This causes a strong interaction with a magnetic field (called ferromagnetism). The three elements that are ferromagnetic are iron, nickel and cobalt. **Figure 4** shows iron nails and the long-range alignment of unpaired electrons that leads to the strong magnetic behaviour of elemental iron.

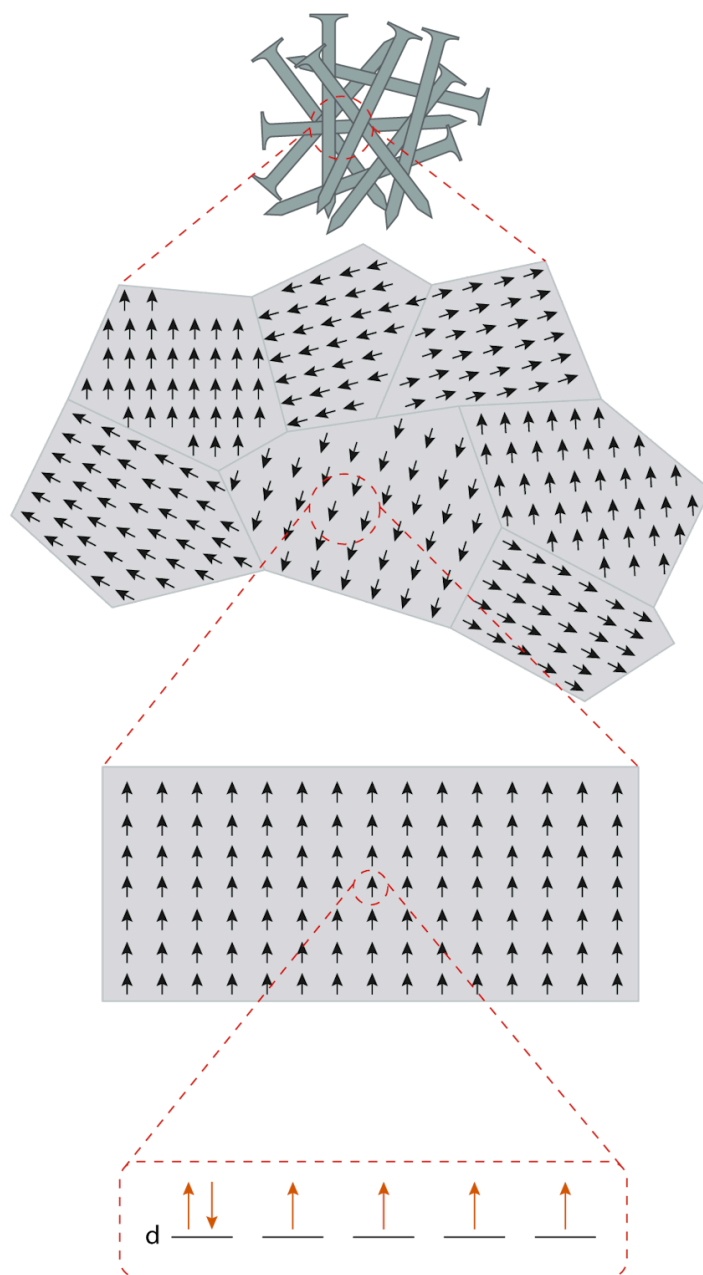


Figure 4. Strong magnetic behaviour of iron.

 More information for figure 4

The image is a diagram illustrating the magnetic alignment in iron. At the top, it shows a collection of gray iron nails grouped together. Below the nails, the diagram zooms in to reveal the atomic structure, represented as a series of pentagonal and hexagonal patterns filled with arrow symbols. These arrows indicate the direction of alignment of the unpaired electrons, demonstrating regions of magnetic domains. A closer view at the bottom part of the diagram focuses on a rectangular section of aligned arrows, indicating ferromagnetic alignment. The detailed inset at the bottom magnifies this alignment, showing arrows pointing mostly upwards, signifying parallel alignment contributing to magnetism. Red dashed lines are used to indicate transitions and focuses on different levels of the illustration.

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Study skills

You do need to be familiar with the concept that transition elements have magnetic properties, but the different types of magnetism (paramagnetic or ferromagnetic) will not be assessed.

4. Catalytic properties

Have you heard the term catalyst before? If so, where have you heard it used? You may be more familiar with enzymes, biological catalysts. Do you know what enzymes do? Catalysts are substances that speed up chemical reactions while remaining unchanged themselves. You will learn about these in more detail in section R2.2.2-3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/factors-affecting-the-rate-of-reaction-id-44714/>).

Transition metals and transition metal ions are often used as catalysts in chemical reactions. Transition metals can provide a surface for the reaction to take place and ions lose or gain electrons to form variable oxidation states. Ions of transition elements can react with one reactant (reactant A) to increase its oxidation state while, afterwards reacting with the other reactant (reactant B) returning to its original oxidation state. The two reactants have either gained or lost electrons allowing them to easily react with one another, forming a product (product C).

Figure 5 shows this in a simplified version of a catalytic cycle diagram for transition metals.

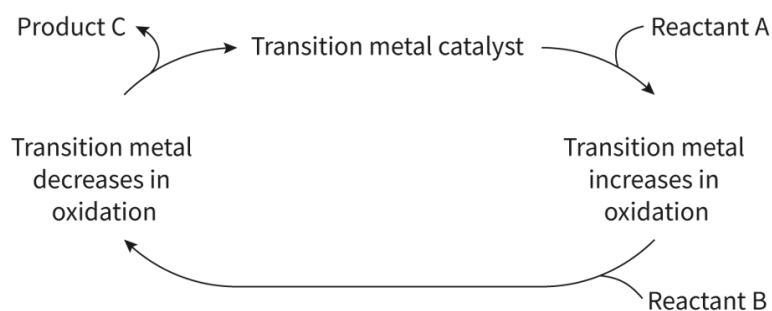


Figure 5. General catalytic cycle diagram for transition metals.

 More information for figure 5

The diagram illustrates a general catalytic cycle for transition metals used in chemical reactions. The cycle begins with Reactant A, which interacts with a transition metal catalyst, increasing the oxidation state of the metal. The transition metal then interacts with Reactant B, allowing the metal to return to its original oxidation state. Product C is formed as a result of this cycle.

Key elements of the diagram include: - "Reactant A" is positioned at the top right and points to the "Transition metal catalyst." - The "Transition metal increases in oxidation" label is on the right side, pointing downward to "Reactant B." - "Reactant B" is at the bottom of the cycle, pointing to "Transition metal decreases in oxidation" on the left. - "Product C" is at the top left, completing the cycle back to the "Transition metal catalyst."

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Since the transition metal is in its original form with its original oxidation state at the end of the reaction, it is not considered a reactant, but rather a catalyst. It simply facilitated the reaction without experiencing a net change.

Haemoglobin is a protein responsible for delivering oxygen in the bloodstream. The haem group within haemoglobin contains the Fe^{2+} ion at its centre. Fe^{2+} can bind to oxygen undergoing a change in oxidation state to Fe^{3+} . When it releases oxygen, it returns to Fe^{2+} . This is illustrated in **Figure 6**.

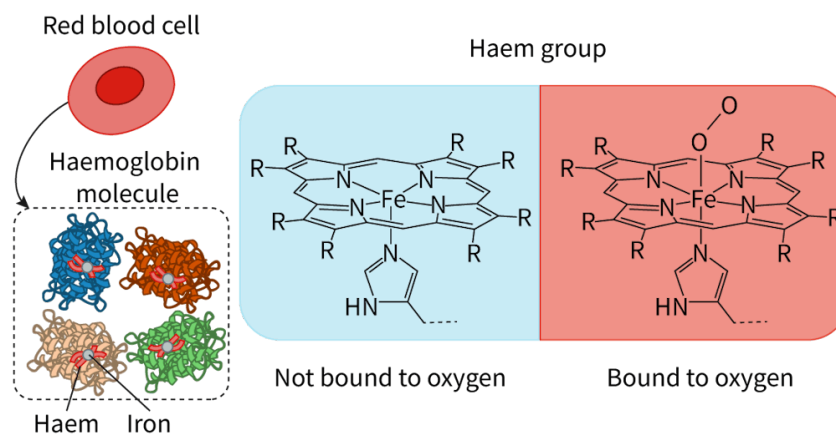


Figure 6. Iron in haem changing oxidation state to bind to iron.

[More information for figure 6](#)

The diagram illustrates the structure of the haem group within haemoglobin, highlighting changes in the iron (Fe) oxidation state. On the left side, the structure is shown without oxygen, where iron is in the Fe^{2+} state. It consists of a porphyrin ring structure bonded with iron at the center, depicted in a blue background. On the right side, the structure is shown bonded with an oxygen molecule, indicating the oxidized Fe^{3+} state, depicted in a red background. The transition between these states is crucial for oxygen binding and release in the bloodstream.

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5. Formation of coloured compounds

Have you encountered any vibrant coloured solutions in chemistry or science so far? Do you remember what solute made up the solution? Chances are you have encountered a coloured solution like the vibrant blue copper(II) sulfate or purple potassium permanganate. These are both shown in **Figure 7**.



Credit: GIPhotoStock, Getty Images

(<https://www.gettyimages.com/detail/photo/pipetting-sodium-carbonate-to-copper-sulfate-both-royalty-free-image/565785491>)



Credit: GIPhotoStock, Getty Images

(<https://www.gettyimages.com/detail/photo/erlenmeyer-flask-containing-a-solution-of-potassium-royalty-free-image/702545749>)

Figure 7. Solutions of copper(II) sulfate (left) and potassium permanganate (right).

Transition elements form coloured compounds. The colours can vary depending on the transition element, the oxidation state, the other species making up the compound, etc. **Figure 8** shows some common transition metal ions and the colour that they typically appear. Notice how Sc^{3+} and Zn^{2+} are colourless. We will look more closely at the theory behind why solutions with transition elements are coloured in the next [section S3.1.10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coloured-complexes-of-transition-element-hl-id-44716/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coloured-complexes-of-transition-element-hl-id-44716/>).

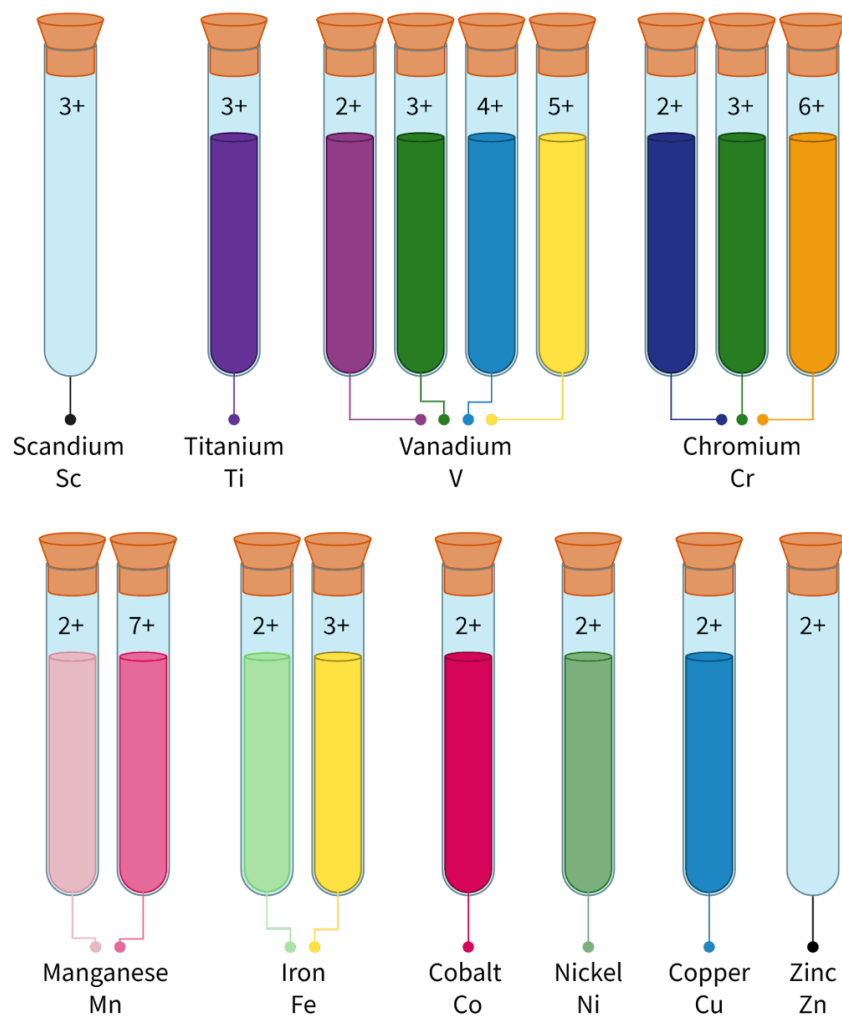


Figure 8. Coloured solutions of some common ions of transition elements.

More information for figure 8

The image depicts an array of test tubes filled with solutions of different colors, each labeled with a charge representing various transition metal ions. The tubes are organized in two rows.

In the first row: 1. A colorless solution with a "3+" label. 2. A purple solution with a "3+" label. 3. A purple solution with a "2+" label. 4. A green solution with a "3+" label. 5. A blue solution with a "4+" label. 6. A yellow solution with a "5+" label. 7. One more purple solution with a "2+" label. 8. A green solution with a "3+" label. 9. A blue solution with a "6+" label.

In the second row: 1. A pink solution with a "2+" label. 2. A dark pink solution labeled "7+". 3. A light green solution with a "2+" label. 4. A yellow solution labeled "3+". 5. A reddish-pink solution with a "2+" label. 6. A green solution labeled "2+". 7. A blue colored solution with a "2+" label. 8. A colorless solution with a "2+" label.

The labels denote different oxidation states of the ions, and their colors indicate the typical appearance in solution as referenced in discussions about transition metal elements and their compounds.

[Generated by AI]

6. Formation of complex ions

We mentioned earlier that the name 'transition' comes from the fact that these elements show transitional behaviour, somewhere between elements in the s-block and p-block. The bonding of transition metal ions to other substances is one important reason why these elements show transitional behaviour. **Figure 9** shows what structure is formed when a Cu^{2+} ion forms a complex with water. Take a look at the structure drawn. What do you notice? Did you expect the structure to look like this? What type of bonds do the lines drawn indicate? Would you expect that type of bond for a transition element or its ion?

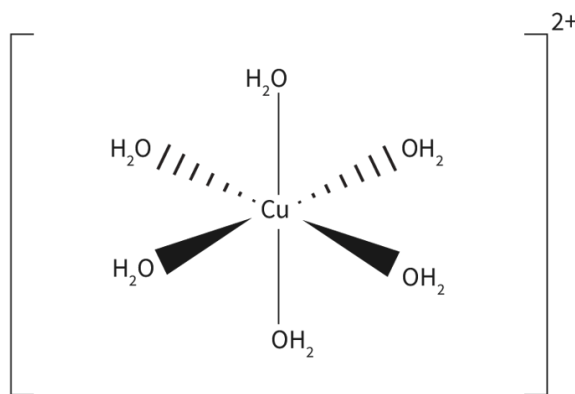


Figure 9. Complex ion of Cu^{2+} and water.

 More information for figure 9

The image shows a diagram of a complex ion where a central copper ion (Cu^{2+}) is surrounded by water molecules. The copper ion is at the center and is connected to six water molecules (H_2O) arranged in an octahedral configuration. The diagram uses solid lines and wedge shapes to indicate the spatial arrangement of the molecules. Each water molecule donates a lone pair of electrons to the copper ion, forming coordination bonds. The copper ion is depicted in the center with a positive charge indicated by the "2+" sign on the right side. The water molecules are labeled with "H₂O" and are drawn with different orientations to illustrate their spatial arrangement around the copper ion.

[Generated by AI]

To form this complex, the lone pairs on the water molecules attract to the positively charged ion and donate a lone pair of electrons to Cu^{2+} forming a coordination bond. You learned about coordination bonds in section S2.2.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>). Water does this multiple times with Cu^{2+} forming a complex ion. This is illustrated in **Figure 10**.

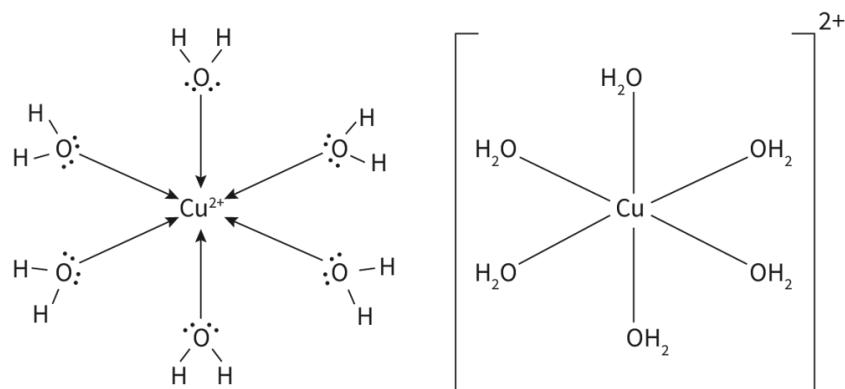


Figure 10. The formation of coordination bonds in the complex ion of Cu^{2+} and water.

 More information for figure 10

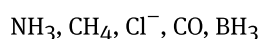
The image shows a diagram illustrating the formation of coordination bonds between a copper ion (Cu^{2+}) and water molecules. On the left side, there is a central Cu^{2+} ion surrounded by six water (H_2O) molecules. Each water molecule is represented with an oxygen atom, depicted with two lone pairs of electrons, bonding with the copper ion. The oxygen atoms point towards the Cu^{2+} ion with arrows indicating the direction of electron donation. The image displays a geometric structure where the water molecules are positioned symmetrically around the Cu^{2+} ion, suggesting an octahedral geometry typical of such complexes. On the right side, the diagram simplifies the complex ion structure, showing the copper ion at the center, surrounded symmetrically by water molecules labeled as H_2O , with lines connecting them to the copper, all enclosed within brackets denoting the complex ion state and the $2+$ charge indication on the top right corner.

[Generated by AI]

The surrounding species that donates a lone pair of electrons to form a coordinate bond with a transition metal ion is known as a ligand. In the previous example, water acted as a ligand by donating one of its lone pairs of electrons to the Cu^{2+} ion. A species can only act as a ligand if it has a lone pair of electrons on at least one atom in a molecule.

Worked example 1

Identify which of the following species can act as a ligand.



A species can only act as a ligand if it has a lone pair of electrons on at least one atom.

If you are not able to identify which substances have lone pairs or not, you will need to draw a Lewis formula.

NH_3 , Cl^- and CO all have at least one lone pair and therefore can act as a ligand.

CH₄ and BH₃ do not have any lone pairs and therefore cannot act as a ligand.

Making connections

The formation of complexes involves a ligand donating an electron pair and a transition metal ion accepting an electron pair in order to form a coordinate bond. In [section R3.4.6](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/lewis-acids-and-bases-id-44715/>), you will learn how species that are electron-pair donors act as a specific type of base known as a Lewis base and how species that are electron-pair acceptors act as a specific type of acid known as a Lewis acid. This means that ligands act as a Lewis base and transition metal ions as a Lewis acid during the formation of a complex.

We can write out the chemical formula of complex ions by using square brackets around the entire complex ion and round brackets around the ligand. For example, in the complex ion we have looked at here, we could write the chemical formula as [Cu(H₂O)₆]²⁺. This shows the overall charge of the ion, the six bonded water ligands and the transition metal.

Worked example 2

Deduce the oxidation state of the transition metal ion and the charge on the complex ion in [Cr(NH₃)₄Cl₂]Br.

In a complex ion, the entire complex is always enclosed inside of the square brackets. Since there is another element written after the square brackets, this would indicate this is an ionic compound. The positive ion would be the complex, with the contents outside the brackets representing the negative ion. We know that bromine (Br) is in group 17 and likely to form the Br[−] ion. This means that the positive ion must have the 1+ charge in order for the ionic compound to be neutral overall.

This means the complex ion is [Cr(NH₃)₄Cl₂]⁺ with a charge of 1+.

To deduce the oxidation state on chromium we now need to look at the ligands and their charge. There are four NH₃ ligands and two Cl[−] ligands. We can create an algebraic equation using x to represent the oxidation state of Cr and summing up the charges of each ligand to equal 1+, the charge of the complex ion.

$$\begin{aligned}x + 4(0) + 2(-1) &= +1 \\x &= +3\end{aligned}$$

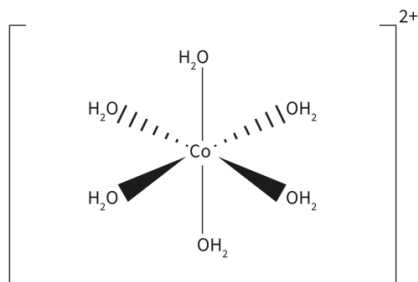
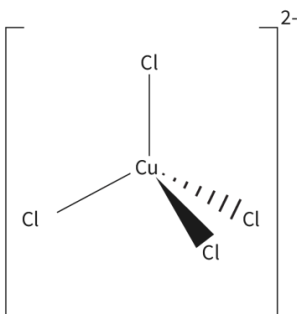
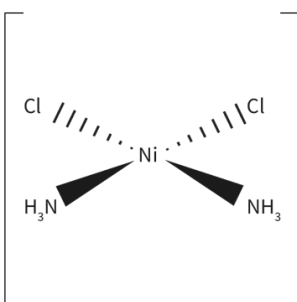
Therefore, the oxidation state on Cr is +3.

In this example we have looked at six bonded ligands, each with one coordinate bond, giving a coordination number of 6. This is a common arrangement of ligands around a transition metal ion, resulting in an octahedral geometry. Other coordination numbers and geometries also exist. **Table 3** shows common coordination numbers and shapes of transition metal complexes.

Table 3. Common shapes of transition metal co

C
o
m
p
l
e
x

Structure

[C
o
(
H₂
O
)
₆]
2+
[More information](#)
[C
u
Cl
₄]
2-
[More information](#)
[N
iCl
₂(
N
H₃
)₂]
[More information](#)

Complex	Structure
$[\text{Ag}(\text{NH}_3)_2]^+$	$\left[\text{H}_3\text{N} - \text{Ag} - \text{NH}_3 \right]$ More information

It is also possible for one ligand to form multiple coordinate bonds with the transition metal ion. This requires the ligand to have more than one lone pair available for coordinate bonding. Two common ligands that form multiple coordinate bonds with transition metal ions are ethylenediamine and ethylenediaminetetraacetic acid (EDTA). **Figure 11** shows two octahedral complexes formed from formation of coordinate bonds with these two ligands. Notice that one ethylenediamine ligand forms two coordinate bonds, whereas EDTA forms six. Three ethylenediamine ligands, each with two coordinate bonds, forms a complex with a coordination number of 6. Only one EDTA ligand bonded will have a coordination number of 6.

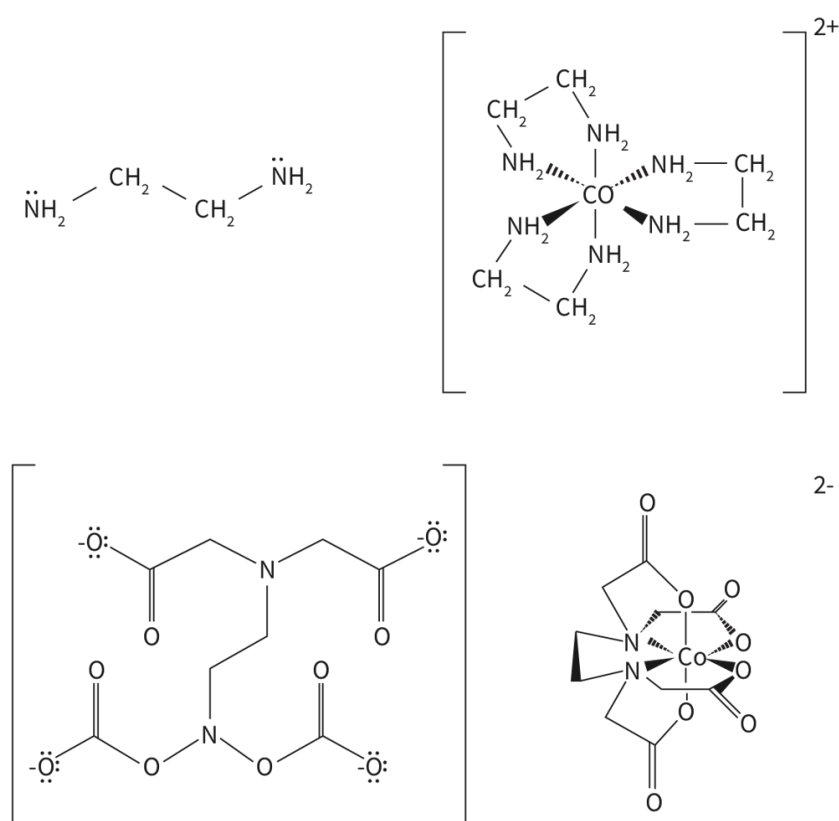


Figure 11. Cobalt complexes with ligands that can form more than one coordinate bond.

 More information for figure 11

The image presents two cobalt complexes with different ligands showing multiple coordinate bonds. The top part shows the structure of ethylenediamine as a ligand, labeled with nitrogen atoms (NH₂) linked by carbon atoms (CH₂). Beside it, a cobalt complex is illustrated, where three ethylenediamine ligands coordinate with a central cobalt ion, resulting in a coordination number of 6 with the notation '2+'.

In the bottom part, the structure of the EDTA ligand is shown with its extended molecular configuration, featuring multiple nitrogen and oxygen atoms set to form coordinate bonds. Adjacent to it, a cobalt complex illustrates how one EDTA ligand bonds with a central cobalt ion, also resulting in a coordination number of 6, indicated by the notation '2-'.

The image illustrates the difference in binding when using ethylenediamine versus EDTA, highlighting how many coordinate bonds each ligand can form with the cobalt ion.

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International Mindedness

EDTA and its ability to form six coordinate bonds to metal ions has extremely useful applications, particularly in medicine. EDTA is used around the world to treat heavy metal poisoning. Ingestion of food or water that has been contaminated with toxic heavy metal ions such as lead can be lethal. Physicians commonly treat patients who have high levels of heavy metal ions in their bloodstream with EDTA. The EDTA binds with heavy metal ions it encounters, preventing the ion from reacting with other substances in the body. After a few hours, the EDTA — metal complex is excreted in urine and is removed from the body. The process of using EDTA medicinally is often referred to as 'chelation therapy'. The word chelate is derived from the Greek word for claw and refers to any ligand capable of forming more than one coordinate bond with a metal ion, encasing the ion like a claw.

While the formation of complex ions is not exclusive to, it is very common in transition elements and they can form a wide variety of complexes. This is why the formation of complexes is considered an important characteristic of transition elements.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills — Being curious about the natural world
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Individual activity

Part 1: Transition metal crossword

Individually, complete the following transition metal crossword. While you should try to avoid it, you may need to research some specific information about a few of the metals.

**Across**

3 Disputes between scientists about whether this element should be a transition metal or not (8)

6 Common metal catalyst used in industry, used for the hydrogenation of oils to make spreadable fats such as margarine (6)

7 Primary metal in steel and ferromagnetic (4)

8 Many uses in the medical field for implants for joint

Check

Show solution

Retry

Interactive 1. Transition Metal Crossword.

 More information for interactive 1

The interactive contains a crossword puzzle with Across and Down clues related to transition metals. The crossword puzzle shuffles on refreshing, and the arrangement of the puzzle and clues varies but the clue and answer remain the same. The grid is arranged with 10 clues spilt into Across and Down. The number of Across and Down clues varies on shuffling. The clues given on the right side are:

Ferromagnetic and commonly forms orange-red complexes with 6 letters,

Many uses in the medical field for implants for joint and bone replacement with 8 letters,

Not technically a transition metal with 4 letters,

Has a melting point just slightly lower than vanadium with 8 letters,

Forms a blue complex in water with 6 letters,

Disputes between scientists about whether this element should be a transition metal or not with 8 letters,

First five successive ionization energies are relatively close together with 8 letters,

Five unpaired electrons with 9 letters,

Common metal catalyst used in industry, used for the hydrogenation of oils to make spreadable fats such as margarine with 6 letters,

Primary metal in steel and ferromagnetic with 4 letters.

The crossword grid has white squares for input and dark blue background squares separating each word, with numbered boxes indicating clue positions. The interface includes buttons labelled Check, Show solution, and Retry. The Check button displays the data as follows.

Read below for solution:

Ferromagnetic and commonly forms orange-red complexes with 6 letters: Cobalt

Many uses in the medical field for implants for joint and bone replacement with 8 letters: Titanium

Not technically a transition metal with 4 letters: Zinc

Has a melting point just slightly lower than vanadium with 8 letters: Chromium

Forms a blue complex in water with 6 letters: Copper

Disputes between scientists about whether this element should be a transition metal or not with 8 letters: Scandium

First five successive ionization energies are relatively close together with 8 letters. Vanadium

Five unpaired electrons with 9 letters: Manganese

Common metal catalyst used in industry, used for the hydrogenation of oils to make spreadable fats such as margarine with 6 letters: Nickel

Primary metal in steel and ferromagnetic with 4 letters: Iron

Part 2: Transition elements used in industry

Pick one transition metal from the crossword and research an industrial process that the transition element is used for. Research the following:

- The catalyst (pure metal or compound)
- The industrial process that uses the catalyst
- Why is the industrial process important for society?

Share your research with a peer.